

PAPER_1574 — $R_{\odot}/R_{\oplus}$ Ratio = $R_{\odot}/R_{\oplus} = SO_5^2 + N_{CH}$ = 100 + 9 = 109

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astronomy **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S577 (Astronomy Unified Proof Set) lists $R_{\odot}/R_{\oplus}$ Ratio as an EXACT-tier closure:

$$R_{\odot}/R_{\oplus} = SO_5^2 + N_{CH} = 100 + 9 = 109$$

UQFF Closed Identity

$$R_{\odot}/R_{\oplus} = SO_5^2 + N_{CH} = 100 + 9 = 109 \quad \text{EXACT}$$

Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Z_S577
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "r_sun_over_r_earth_109"})`

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